Team Details

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Problem Statement

Selected Problem

Real-Time Multi-User Smart Assistant for Dynamic and Noisy Smart Environments

Our Problem Understanding (Use Cases Addressed)

1. The system must **separate overlapping speech from a single input** and identify who is speaking in real time. It should handle simultaneous voices, map them to user profiles, and filter out disturbances.
2. Recognize and accurately **detect multiple keywords** **spoken simultaneously** within overlapping audio streams.
3. After separating voices, the **system must decide: Is this a command, conversation, noise, new convo. or existing one ?.**This is where most assistants fail **(false triggers and wrong actions)**.
4. When multiple valid commands appear, the system should **resolve conflicts intelligently** and know when to ask for clarification.
5. Personalization must be accurate and context-aware, while still keeping **resource usage efficient**.
6. Existing assistants treat every audio event with **equal computational weight**. **The real challenge** is **knowing which audio events deserve computation at all and escalating resources** only when the scene genuinely demands it.

Proposed Solution

Our Solution – The Canary

"Because not every sound deserves a million parameter computation. Every sound must first earn its compute."

The Passive Idle Gate

Use Cases Addressed ***6,1***

A lightweight layer silently guards the entire pipeline. It outputs just three signals: **speech_detected | wakeword_probability | noise_floor.** If nothing clears the threshold the system stays idle.

Shown in Stage - 0

Dynamic Routing

Use Cases Addressed ***3,5***

Every acoustic event receives a **Scene Complexity Score (SCS)** based on **speech probability, overlap severity, wake-word certainty, and contextual relevance. Only high-scoring events escalate .**

Shown in Stage - 1, Dynamic Resource Scaler

Multi Stage Audio Intelligence

Use Cases Addressed ***2,3***

A dedicated **lightweight layer estimates speaker** count, overlap probability, & directed speech relevance before any separation model is triggered. It then assigns scores based on **speaker prioritization**.

Shown in Stage - 1, 2

Uncertainty-Aware Execution

Use Cases Addressed ***4***

Every module outputs 3 values: **predicted intent, confidence score, signal quality metrics**. The system then decides execute, ignore, or ask for clarification based on accumulated info. across the pipeline. **It never blindly trusts its own output.**

Shown in Stage - 2

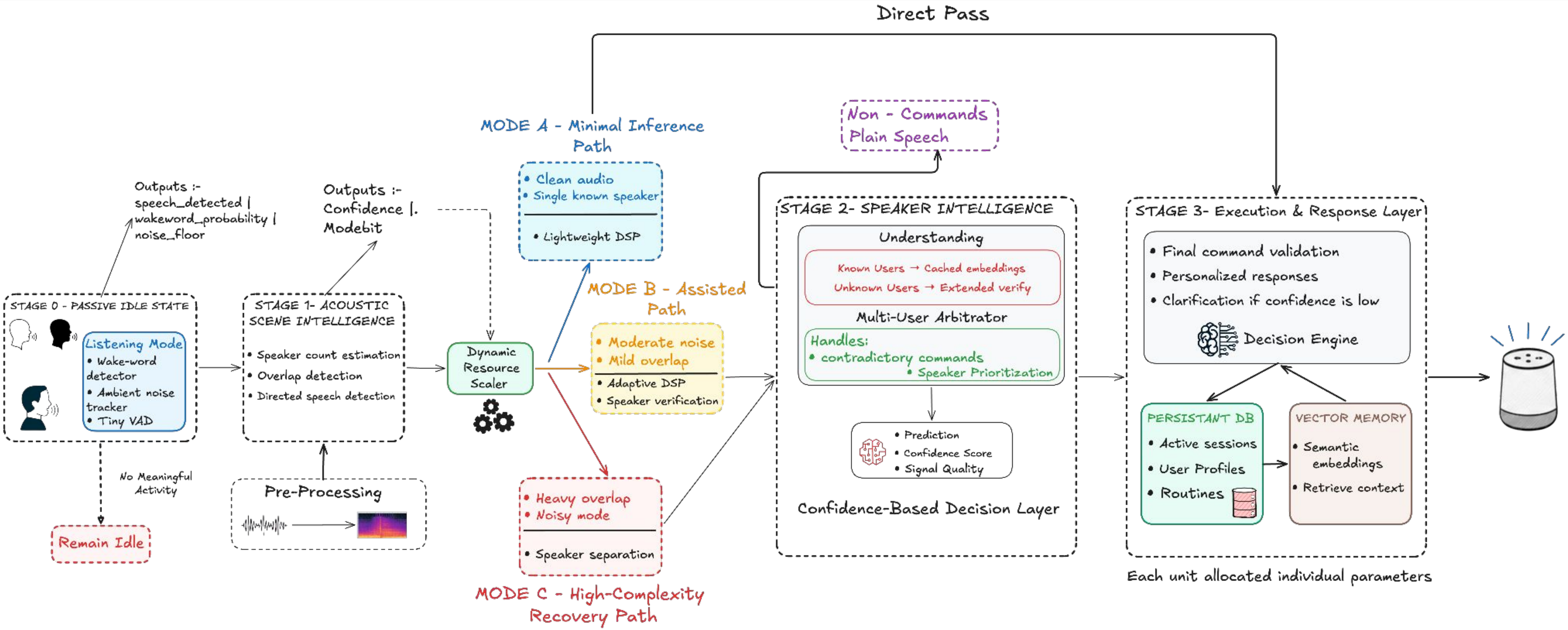
Sequential Execution

Use Cases Addressed ***2,6***

Commands are executed using a **priority_queue** based on confidence, speaker identity, and overlap severity. Ambiguous or conflicting requests are escalated for clarification.

Shown in Stage - 3

Proposed Solution – Overview Of The Entire Technical Architecture



The Canary starts with a **Passive Idle State** and **a listening mode** that checks wake-word confidence, speech activity, overlap, and noise before activating computation. A **Dynamic Resource Scaler** then routes audio through **Mode A**, **B**, or **C** depending on scene complexity, after which the **Speaker Intelligence** and **Confidence-Based Decision Layer** handle speaker verification, conflict resolution, and uncertainty scoring. Finally, the **Execution & Response Layer** combines **data stored & personalization context** to deliver real-time, and multi-user aware smart home responses.

Proof of Working: Step-by-Step Logic

1. Passive Idle State Logic (Wake-word Gate)

$$\text{PASS} = \underbrace{\mathbb{1}[P_{ww} > \tau_1]}_{\text{wake-word}} \cdot \underbrace{\mathbb{1}\left[\frac{1}{N}\sum x_i^2 > \tau_2\right]}_{\text{frame energy}} \cdot \underbrace{\mathbb{1}[\text{SNR} > \tau_3]}_{\text{noise floor}}$$

All conditions must clear simultaneously:

- Wake-word confidence > τ1
- Frame energy > τ2
- Signal-to-noise ratio > τ3

If any fail, the system stays completely idle.

2. Speaker Separation & Identification

$$\text{SI-SNR} = 10\log_{10}\frac{\|\alpha s\|^2}{\|\hat{s} - \alpha s\|^2}, \qquad \alpha = \frac{\hat{s}^\top s}{\|s\|^2}$$

$$\text{Accept speaker} = \mathbb{1}\left[\frac{e_1 \cdot e_2}{\|e_1\|\,\|e_2\|} > \delta\right]$$

- **SI-SNR** measures separation quality (reference vs. output)
- **α** removes scale ambiguity for shape comparison
- **Cosine similarity** vs. stored profiles confirms user (threshold δ)

3. Dynamic Resource Scaler

$$\text{SCS} = w_1 O + w_2 N + w_3 P \qquad \sum w_i = 1$$

where

O = overlap probability

N = normalized noise level

U = speaker uncertainty score

- **Pww | Poverlap | Ds:** Score audio before heavy execution
- **Low Score** ⟶ Fast lightweight path
- **High Score** ⟶ Full separation and denoising stack

Compute only activates when the scene demands it.

4. Confidence-Based Decision Layer

$$U_{\text{total}} = 1 - \prod_{i=1}^{N}(1 - u_i), \qquad M_i = (\hat{y}_i,\ c_i,\ u_i), \qquad c_i + u_i \le 1$$

$$\text{Action} = \begin{cases} \textit{Execute} & c_i > 0.80 \ \wedge \ U_{\text{total}} < 0.20 \\ \textit{Escalate} & 0.50 < c_i \le 0.80 \\ \textit{Clarify} & c_i \le 0.50 \end{cases}$$

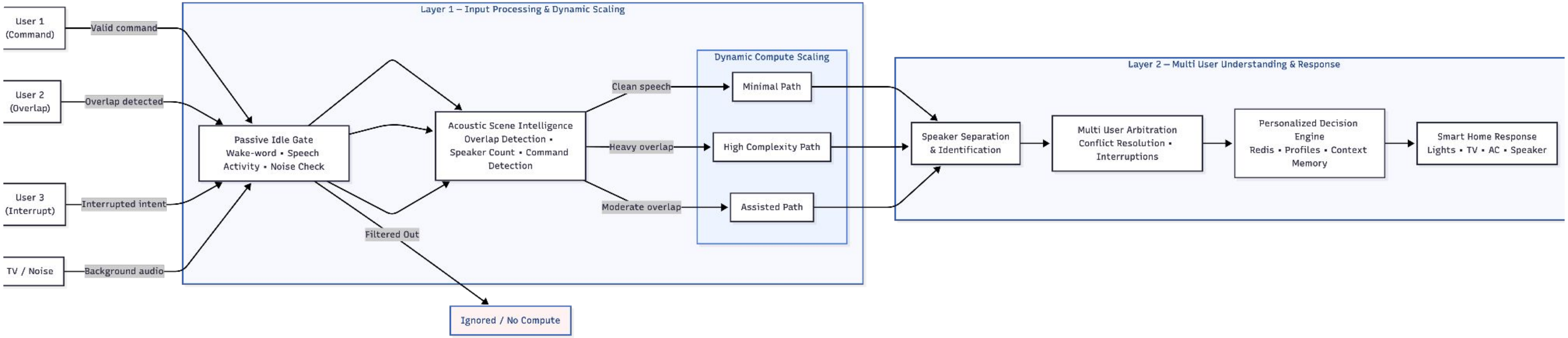
- **Every module in the pipeline votes** with 3 numbers ⟶ What it thinks happened, How sure it is, How unsure it is
- **Only when the system is confident enough across all stages** it actually executes a command, otherwise it asks you to repeat or clarifies first.

Proposed Solution – How We Address Each Problem Statement ?

PROBLEM STATEMENT	HOW THE CANARY ADDRESSES IT	TARGET FOCUS
Real-Time Multi-Speaker Separation	Uses a **staged modular pipeline** where heavier models activate by **Dynamic Scaler** only when there is overlap and speech detected else **lighter** filtering are used	<5M Parameters, xRT < 0.5
Separating Multiple Voices	Estimates **speaker count**, **overlap**, and noise level first, then selects the required **separation path** to isolate **overlap**ping voices more accurately.	High SI-SNR, Cleaner Separation
Distinguishing Commands vs Noise	A **Passive Idle Gate** checks **wake-word confidence**, **speech activity**, and room noise so random conversations or TV audio do not activate the assistant.	False Trigger Reduction
Turn-Taking & Overlapping Speech	Tracks **active speakers**, **interruptions**, and **conversational flow** in real time using to understand **simultaneous commands** and evolving intent.	Speaker Tracking, Parallel Handling
Multiple Keyword Recognition	Processes each separated **speaker stream independently** so **multiple commands** and keywords can be detected without mixing contexts.	Low WER During Overlap
Personalized Responses	Maps identified voices to **user profiles**, routines, and permissions using **lightweight caching** and **memory systems** for fast contextual responses.	Fast Context Retrieval

UML Diagram, Plausibility & Constraints

System Architecture & UML Logic Flow



PLAUSIBILITY & TECHNICAL CONSTRAINTS

Speech Separation

Adopt **Samsung TFPSNet** for real-time overlapping speaker separation (>21 dB SI-SDRi) **within a 5M-parameter budget.**

Wake-Word & Noise

Integrate INT8-quantized Silero VAD and WakeWord CNN with MatchboxNet backbone for **intent detection and ambient noise rejection.**

ASR & Speaker ID

Use **ASR fine-tuned on IndicVoices** with the CAM++ B0 model for secure **speaker verification and identity tagging.**

Local Deployment

Build a **fully local, cloud-free pipeline** using a **Python arbitration engine** with weighted scoring and SmartSense logs for **conflict resolution.**

Novelty in our approach (Drawing Comparisons b/w Contemporary & Canary)

Conventional Methods

Cloud Based

Most assistants rely primarily on **cloud-first processing**, meaning audio is streamed across, increasing latency and dependency on internet .

False Wake-Up's

Some assistants can still suffer from **false wake-ups** caused by conversations, or nearby devices because wake-word activation is always listening.

More Compute

Current systems mainly process audio through a **larger unified pipeline**, even for simple commands, increasing unnecessary compute usage.

The Canary

Lightweight

The Canary uses a lightweight modular pipeline with local caching, and resource allocator so only the required modules activate .

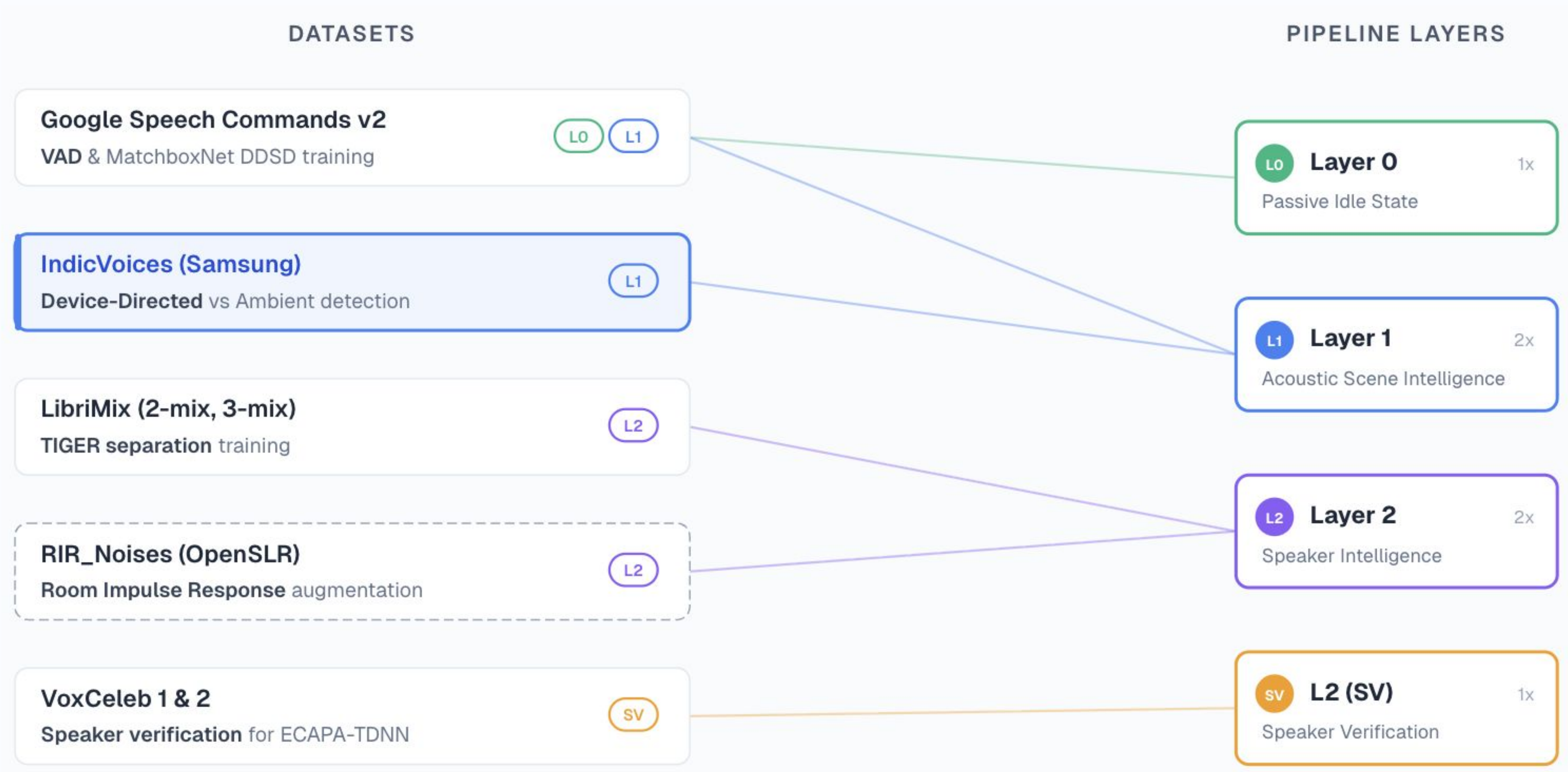
Passive Filtering

A Passive Idle Gate continuously checks wake-word confidence, speech activity, noise before allowing further computation, filtering out low-worthiness events early.

Dynamic Scaling

The Canary uses Scene Complexity Score that routes audio through Minimal, Assisted, or High-Complexity paths depending on overlap, noise, and scene complexity.

Open Datasets planned to be used



Open Models planned to be used, developed, trained, fine-tuned

Model	Role	Params	Pipeline Stage
Layer 0: Passive Idle State			
Silero VAD INT8	Voice **activity detection** + energy floor estimation. Gates all downstream processing.	~0.5M	Continuous listen → Silent (stop) or Activity (→ Layer 1)
openWakeWord	**Wake-word detection** after VAD confirms non-silence.	~0.4M	Wake confirmed → fast-path to Layer 1 · not-wake → stop
Layer 1: Acoustic scene intelligence			
MatchboxNet 3×2×1	**Background noise rejection**. Noise class + overlap probability estimation.	~0.093M	Single-speaker (~85%) vs Overlapping audio (~15%) routing decision
Layer 2: Speaker Separation			
Samsung TFPSNet	**Primary multi-speaker separation**. Interleaved time-frequency attention + band-splitting.	~2.6M	Mode C: High-complexity separation · SI-SNR >18 dB
CAM++ (B0 Variant)	**Speaker verification** + embedding extraction. **FAISS lookup** for known speakers.	~1.1M	Mode B/C: Speaker verification → identity tag to Layer 3
Layer 3: Execution and Response Layer			
Rule-based engine prototype fallback	**Weighted scoring** across 4 confidence distributions. **RBAC** + temporal priority.	0 (Python)	Lightweight arbitration · deterministic, <1 ms

AI Tools Used For Canary

Lightweight Local Decision Engine

Using a lightweight **local SLM with rule-based** orchestration to process commands, handle clarifications, and resolve conflicting requests between multiple users in real time.

End-to-End Smart Workflow

Building an AI-driven response pipeline using speech separation models like **TFPSNet/TIGER,** followed by automated dispatch workflows from noisy overlapping audio.

Centralized Routing & Resource Management

Designing a **central AI orchestration controller** that **dynamically routes audio** through different processing paths based on overlap, noise level, and speech confidence to keep the system lightweight and efficient.

Audio Understanding Before Heavy AI Processing

Using lightweight acoustic DSP & AI models such as **MatchboxNet** for **speech activity, overlap detection, and noise filtering** before activating larger separation or language models, reducing unnecessary compute usage.

Supporting Research Materials

1. Silero VAD - Lightweight **VAD used for speech detection** and passive pipeline gating. github.com/snakers4/silero-vad

Used in: Stage 0 - Passive Idle Gate

2. MatchboxNet - A Neural net used to compute **wake-word confidence** as one of the three gate signals. arxiv.org/abs/2004.08531

Used in: Stage 0 - Wake-Word Detection

3. Conditional Computation in Neural Networks - Bengio et al. Establishes principle of **selectively activating** parts of a model. arxiv.org/abs/1511.06297

Used in: Stage 1 - Dynamic Resource Scaler

4. Dual-Path RNN (DPRNN) - Luo et al. A **time-domain separation model** for multi-speaker isolation. arxiv.org/abs/1910.06379

Used in: Stage 1 Mode C - Heavy Separation Path

5. ECAPA-TDNN - Desplanques et al. **Speaker embedding model** using channel attention and multi-scale aggregation. arxiv.org/abs/2005.07143

Used in: Stage 2 - Speaker Intelligence & Multi-User Arbitration

6. A Survey of Uncertainty in Deep Neural Networks - Gawlikowski et al. **Review of confidence estimation methods**. arxiv.org/abs/2107.03342

Used in: Stage 2B - Confidence-Based Decision Layer

7. LibriMix - Cosentino et al. **Dataset for training separation models** and benchmarking SI-SNR and WER. arxiv.org/abs/2005.11262

Used in: Training & Evaluation